QUANTUM COMPUTING FACTS AND FIGURES

- **Number of Qubits:** As of February 2024, IBM announced a 135-qubit processor, pushing the limit beyond 100. For comparison, a 50-qubit system can process more data than there are atoms in the Earth. However, larger systems don't automatically translate to superior performance.
- **Temperature for Superconducting Qubits:** Superconducting quantum computers operate at temperatures close to absolute zero, around -273.15°C (-459.67°F), colder than outer space.
- **Quantum Supremacy:** Google claimed to achieve quantum supremacy in 2019 with their 54-qubit processor, Sycamore, solving a specific problem in 200 seconds that would take the world's most powerful supercomputer 10,000 years to solve.
- **Ion Trap Precision:** Ion trap quantum computers use lasers to manipulate qubits with precision up to 99.9999%, a key factor for executing quantum algorithms accurately.
- **Decoherence Time:** Quantum systems can maintain their state coherently from microseconds to milliseconds, depending on the qubit technology. Extending decoherence times is a primary focus for researchers.
- **Error Rates:** Current quantum processors have error rates between 0.1% and 1% per quantum gate operation. Quantum error correction codes aim to reduce this to make fault-tolerant quantum computing feasible.
- **Quantum Volume:** A metric introduced by IBM, Quantum Volume measures the overall performance and capability of a quantum computer. It considers factors like qubit count, connectivity, and error rates.
- **Investment in Quantum Computing:** Global spending on quantum computing research and development is estimated to exceed $10 billion annually, with both public and private sector investments.
- **Quantum Algorithms:** Over 50 quantum algorithms have been proposed that could solve certain problems more efficiently than their classical counterparts, ranging from cryptography to materials science.
- **Quantum Internet:** Efforts are underway to develop a quantum internet that would use quantum entanglement for secure communication. The first intercontinental quantum-secured communication was demonstrated between Europe and China.

robust error correction techniques. These techniques aim to identify and correct errors during computations, making the overall results more reliable.

Other Players

- **Rigetti Computing:** Rigetti focuses on hybrid quantum-classical computing, combining quantum processors with classical computers for enhanced efficiency. Their Aspen platform offers access to superconducting quantum devices for specific tasks, targeting practical applications in finance and machine learning.
- **IonQ:** IonQ utilises trapped-ion qubits, offering longer coherence times and potentially higher fidelity compared to superconducting qubits. Their trapped-ion processors currently reach up to 32 qubits and cater to areas like quantum chemistry and optimization problems.
- **D-Wave:** D-Wave offers a unique approach with quantum annealers, specialised processors designed for solving specific optimization problems. While not universal quantum computers, their technology finds applications in logistics, scheduling, and finance.

The diverse approaches among these players highlight the multifaceted nature of quantum computing research. Each company's strengths and focus areas contribute to the broader advancement of the field. Collaborations and open-source initiatives like Qiskit further accelerate progress by fostering knowledge sharing and innovation. d